

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 2

The following four questions are your warm-up exercises. We will not be discussing them during tutorial.

Ex. 6.2.2. Use the (generalized) pigeonhole principle to show that if there are 30 students in a class, then at least two have last names that begin with the same letter.

Additional Question. 250 courses have their final exams scheduled over a period of 20 days. There are 3 possible time slots in which an exam can be held on a given day. At most two exams can take place in the same venue at the same time. Show that at least 3 venues need to be booked.

Ex. 6.3.3. Given $S = \{a, b, c, d, e, f, g\}$, how many 5-permutations end with a ? How many 5-combinations contain a ?

Ex. 6.3.27. How many bit strings of length 10 contain at least three 1's and at least three 0's?

For the tutorial on 1 February, let's discuss the following questions (as many as time permits).

Ex. 6.2.4, 6.2.8, 6.2.32, Additional Question, 6.3.7, 6.3.8, 6.3.13, 6.3.16, 6.3.19, 6.3.25.

Ex. 6.2.4. A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.

- (a) How many balls must she select to be sure of having at least three balls of the same color?
- (b) How many balls must she select to be sure of having at least three blue balls?

Ex. 6.2.8. Let $(x_i, y_i), i = 1, 2, 3, 4, 5$, be a set of five distinct points with integer coordinates in the xy plane. Show that the midpoint of the line joining at least one pair of these points has integer coordinates.

Ex. 6.2.32. Prove that at a party where there are at least two people, there are two people who know the same number of other people there. (Assume knowledge is mutual, e.g., if A knows B, then B knows A as well.)

Additional Question. Assume the relation between any two people is either friend or stranger, prove that among any party of 6 people, there exist three people who are all friends to one another, or all strangers to one another.

Ex. 6.3.7. How many bit strings of length 10 contain

- (a) exactly four 1's?
- (b) at most four 1's?
- (c) at least four 1's?
- (d) an equal number of 0's and 1's?

Ex. 6.3.8. A group contain n men and n women. How many ways are there to arrange these people in a row if the men and women alternate?

Ex. 6.3.13. How many permutations of the letters $ABCDEFGH$ contain

- (a) the string BCD ?
- (b) the string $CFGA$?
- (c) the strings BA and GF ?
- (d) the strings ABC and DE ?
- (e) the strings ABC and CDE ?
- (f) the strings CBA and BED ?

Ex. 6.3.16. Thirteen people on a softball team show up for a game.

- (a) How many ways are there to choose 10 players to take the field?
- (b) Of the 13 people who show up, three are women. How many ways are there to choose 10 players to take the field if at least one of these players must be a woman?

Ex. 6.3.19. How many 4-permutations of the positive integers not exceeding 100 contain three consecutive integers $k, k+1, k+2$, in the correct order

- (a) where they are in consecutive positions in the permutation?
- (b) where these consecutive integer can perhaps be separated by other integers in the permutation?

Ex. 6.3.25. How many bit strings contain exactly eight 0's and 10 1's if every 0 must be immediately followed by a 1?

The questions below are optional. Attempt them to challenge yourself, but we will not be discussing them.

Additional Question. [Hard, IMO 2001] Suppose that 21 girls and 21 boys enter a mathematics competition. Furthermore, suppose that each entrant solves at most 6 questions, and for every boy-girl pair, there is at least 1 question that they both solved. Show that there is a question that was solved by at least 3 girls and at least 3 boys.

Additional Questions on Poker Cards. Five card poker is good setting in which we can practice our counting skills. In case you have never been to a casino, a card deck consists of 52 cards; each card has one of the 13 face values 2 3 4 5 6 7 8 9 10 J Q K A and one of the four suits ♣, ♦, ♥, ♠. In five card poker, you are dealt a hand consisting of five different cards, for example

$$2♠, 9♥, K♦, A♥, 2♥$$

and you win a prize if your hand is of a special type. We'll apply counting rules to figure out the number of different types of hands.

Four-of-a-kind. A hand is a *four-of-a-kind* if it contains four cards with the same value, for example:

$$\{J♥, J♦, J♣, J♠, 2♦\}.$$

Full house. A hand is a *full house* if it consists of three cards with one face value and two cards with another face value, for example,

$$\{8♠, 8♦, 8♣, Q♦, Q♠\}.$$

Flush. A hand is a *flush* if all five cards are of the same suit,¹ for example,

$$\{4♣, A♣, 9♣, K♣, 7♣\}.$$

Two-pair. A hand is a *two-pair* if it has two cards of one face value, two cards of another face value, and a fifth card of yet another face value, for example,

$$\{A♠, A♣, 8♦, 8♣, 9♥\}.$$

Note that the order of the cards does not matter in a hand.

- (a) How many *four-of-a-kind* hands are there?
- (b) How many *full house* hands are there?
- (c) How many *flush* hands are there?
- (d) How many *two-pair* hands are there?

Additional Question. Use the inclusion-exclusion principle to answer the following question: How many permutations of the 10 letters $\{a, b, c, d, e, f, g, h, i, j\}$ are there that do not contain any of the patterns *hi*, *fad*, and *jig*? For example, the permutations *hedafjigcb* and *jghiebfadc* should not be counted because the first one contains *jig* and the second one contains both *hi* and *fad*.

¹In poker there is also a “royal flush” and a “straight flush” that are sometimes counted separately. We will include those into our count of flushes.